

Modelling photodissociation regions with the 3D-PDR code

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Editor: [✉](#)

Submitted: 10 June 2026

Published: unpublished

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Summary

3D-PDR is an open-source astrochemical code designed to model the chemistry, thermal balance, and line emission of photodissociation regions (PDRs) in one-, and three-dimensional environments. The code solves the coupled interaction between far-ultraviolet, gas-phase chemistry, dust processes, thermal physics, and level populations, enabling the prediction of molecular abundances, temperatures, and observable emission from the interstellar medium. The upgraded version presented here introduces the high-performance ray-tracing algorithm RAYTHEIA, together with new physical modules and a modular software architecture that significantly improve computational efficiency, scalability, and scientific capabilities. The code provides predictions using 1D and 3D density distributions, including post-processing of hydrodynamical simulations and the generation of synthetic observations for comparison with data from current and future instruments.

State of the field

Photodissociation regions are dynamic interfaces in the interstellar medium (ISM) where ultraviolet photons (UV) from nearby massive stars strongly influence the physical and chemical state of the gas and dust (Hollenbach & Tielens, 1999; Tielens & Hollenbach, 1985; Wolfire et al., 2022). PDRs are regions acting as transition zones between the ionized and molecular phases and where the atomic-to-molecular phase is located. They are characterized by the dissociation of molecules, especially that of H_2 , as well as the ionization of atoms, leading to the formation of complex structures and the emission of molecular, atomic and ionic lines. PDRs are found in environments ranging from molecular cloud edges to surfaces of star-forming regions, as well as HII regions. Since the bulk of the neutral ISM emission is associated with PDRs (Hollenbach & Tielens, 1999; Wolfire et al., 2022), understanding them is crucial for studying the small and large scale of the ISM, including the evolution of galaxies.

Numerical modeling of PDRs presents significant challenges, due to the coupling of the given model distribution with radiative transfer and chemical pathways involving hundreds to thousands of reactions in non-equilibrium environments. Various models of PDRs are often one-dimensional, assuming simple slab or spherical geometries (Röllig et al., 2007).

The 3D-PDR code has been one of the earliest attempts in modelling the PDR chemistry in three-dimensional density distributions (T. G. Bisbas et al., 2012). In particular, 3D-PDR can be used as a one-dimensional code for fast PDR calculations (T. G. Bisbas et al., 2014; Thomas G. Bisbas et al., 2015; Gaches et al., 2019a, 2019b; Luo et al., 2023, 2024) and the construction of one-dimensional thermochemical grids covering a wide range of ISM environmental parameters (Thomas G. Bisbas et al., 2019, 2023, 2025; Dasyra et al., 2022;

42 [Gaches et al., 2015](#)). It can also be used as a three-dimensional tool to post-process snapshots
43 of molecular clouds and star-forming regions modelled with hydrodynamical codes ([Thomas G.
44 Bisbas, van Dishoeck, et al., 2017](#); [Thomas G. Bisbas, Tanaka, et al., 2017](#); [Thomas G. Bisbas
45 et al., 2018, 2021, 2024](#); [Gaches, Bisbas, et al., 2022](#); [Gaches, Bialy, et al., 2022](#); [Obolentseva
46 et al., 2024](#)) and to generate synthetic observations for direct comparison with observational
47 data.

48 The upgraded 3D-PDR presented here, offers a significant speed-up in calculations of three-
49 dimensional structures, thanks to the new ray-tracing algorithm RAYTHEIA ([Zhu et al.,
50 2026](#)).

51 Statement of need

52 While computationally efficient with highly sophisticated chemistry, one-dimensional models
53 cannot capture the complexity of three-dimensional structures that real interstellar clouds
54 exhibit and therefore their abundances distribution and emission properties. A fully three-
55 dimensional approach is, therefore, essential to realistically model these environments, allowing
56 researchers to connect theory with the detailed observations now possible with radio-telescopes.

57 The 3D-PDR code provides a state-of-the-art numerical framework to simulate the chemistry,
58 thermal balance, and line emission of PDRs. To our knowledge, 3D-PDR is the only available
59 code for treating PDRs in three-dimensions. The upgraded version of 3D-PDR includes
60 significant improvements in computational performance, modularity, and usability, enabling
61 large-scale parameter studies and integration with hydrodynamical processes. In doing so, it
62 advances the field beyond earlier one-dimensional codes and provides an important tool for
63 researchers seeking to interpret observations in the era of facilities such as ALMA and JWST.

64 Research impact statement

65 The upgraded 3D-PDR code enables three-dimensional astrochemical and radiative-transfer
66 calculations at resolutions that were previously computationally prohibitive. By combining a
67 dual-grid framework, adaptive octree traversal, analytical path-length calculations, and efficient
68 hybrid parallelization, the code can model the chemistry, thermal balance, and line emission of
69 PDRs in high-resolution three-dimensional simulations. This capability bridges the gap between
70 modern large-scale hydrodynamical simulations and detailed astrochemical modelling, allowing
71 the community to study the interaction between radiation and the ISM in full three-dimensions.

72 The software provides a platform for investigating a broad range of astrochemical problems,
73 including molecular cloud evolution, star formation, feedback processes, and the interpretation
74 of observations from flagship facilities such as ALMA and JWST, and future instruments. This
75 makes 3D-PDR a powerful tool for both theoretical studies and the generation of synthetic
76 observations used to connect simulations with observed data.

77 Software design

78 The upgraded 3D-PDR code features the new ray-tracing algorithm RAYTHEIA ([Zhu et
79 al., 2026](#)). RAYTHEIA is a high-performance reverse ray-tracing algorithm designed to
80 efficiently solve three-dimensional direction-dependent equations in astrophysical simulations,
81 with particular focus on photodissociation regions (PDRs). The method adopts a dual-grid
82 framework in which the native simulation mesh serves as the source grid for ray emission, while
83 an adaptive Cartesian contribution grid is used for efficient ray traversal and accumulation of
84 directional quantities. Rays are discretized using the HEALPix scheme ([Górski et al., 2005](#))
85 and treated as infinitesimally thin pencil beams.

Key innovations of RAYTHEIA include a leaf-only linear-octree data structure combined with Morton-code indexing and digital differential analyser (DDA) traversal for efficient ray-walking, an analytical slab method for exact path-length calculations through grid cells, and a hybrid MPI/OpenMP distributed-memory framework employing a chunk-to-chunk communication strategy. These advances provide near-ideal parallel scalability, substantially reduce memory requirements, and enable high-resolution three-dimensional PDR simulations. RAYTHEIA is modular, computationally efficient, and can be readily integrated into existing simulation codes requiring fast solutions of direction-dependent radiative-transfer problems.

3D-PDR performs thermal balance and level populations iterations and terminates once the total heating matches with the total cooling to within a user-defined tolerance parameter. It outputs the abundance distribution of species of the given chemical network, gas and dust temperatures, local heating and cooling for all individual processes, as well as emissivities and level populations of the coolants considered. The latter can be further post-processed to construct synthetic observations by solving the radiative transfer equation from the point of view of the observer.

The code includes several new physical modules from those initially presented in T. G. Bisbas et al. (2012) that extend its applicability to a wider range of interstellar environments. These additions include routines for modelling cosmic-ray energy spectra and their attenuation through gas columns (Gaches et al., 2019a; Gaches, Bisbas, et al., 2022), allowing for a more accurate treatment of cosmic-ray ionization rates. The code also incorporates the suprathermal formation pathway of CO via CH^+ (Thomas G. Bisbas et al., 2019; Visser et al., 2009), which becomes important in regions where non-thermal processes enhance the formation of CO at low column densities. Furthermore, the chemical network has been expanded to include electron-ion recombination on dust grain surfaces (Weingartner & Draine, 2001), providing a better description of charge balance in dense and shielded gas. Finally, the upgraded version introduces the calculation of H_2 ro-vibrational and rotational line emission (Gaches et al., submitted), enabling direct comparison with JWST data.

Example case

As an example, we post-process a sub-region from the SILCC-Zoom hydrodynamical simulation of Seifried et al. (2017) modelling a star-forming molecular cloud. The selected region has a mass of $M_{\text{tot}} = 7.3 \times 10^4 M_{\odot}$ and an extent of 62.5 pc resolved with 256^3 uniform cells achieving a spatial resolution of 0.24 pc. It contains a filamentary structure with significant substructure, including dense clumps and extended low-density regions. We consider an FUV strength of $\chi/\chi_0 = 10$ (Draine, 1978) and the L -function of Padovani et al. (2018) describing the cosmic-ray ionization rate and its attenuation along depth. We assume solar metallicity, dust-to-gas ratio of 10^{-2} and a sub-set of the UMIST2012 chemical network (McElroy et al., 2013) consisting of 79 species and 1169 reactions. The initial elemental abundances were set as follows: $\text{C}^+ = 10^{-4}$, $\text{O} = 3 \times 10^{-4}$, $\text{N} = 3.3 \times 10^{-5}$, $\text{Mg}^+ = 2.7 \times 10^{-7}$, and $\text{He} = 8.5 \times 10^{-2}$.

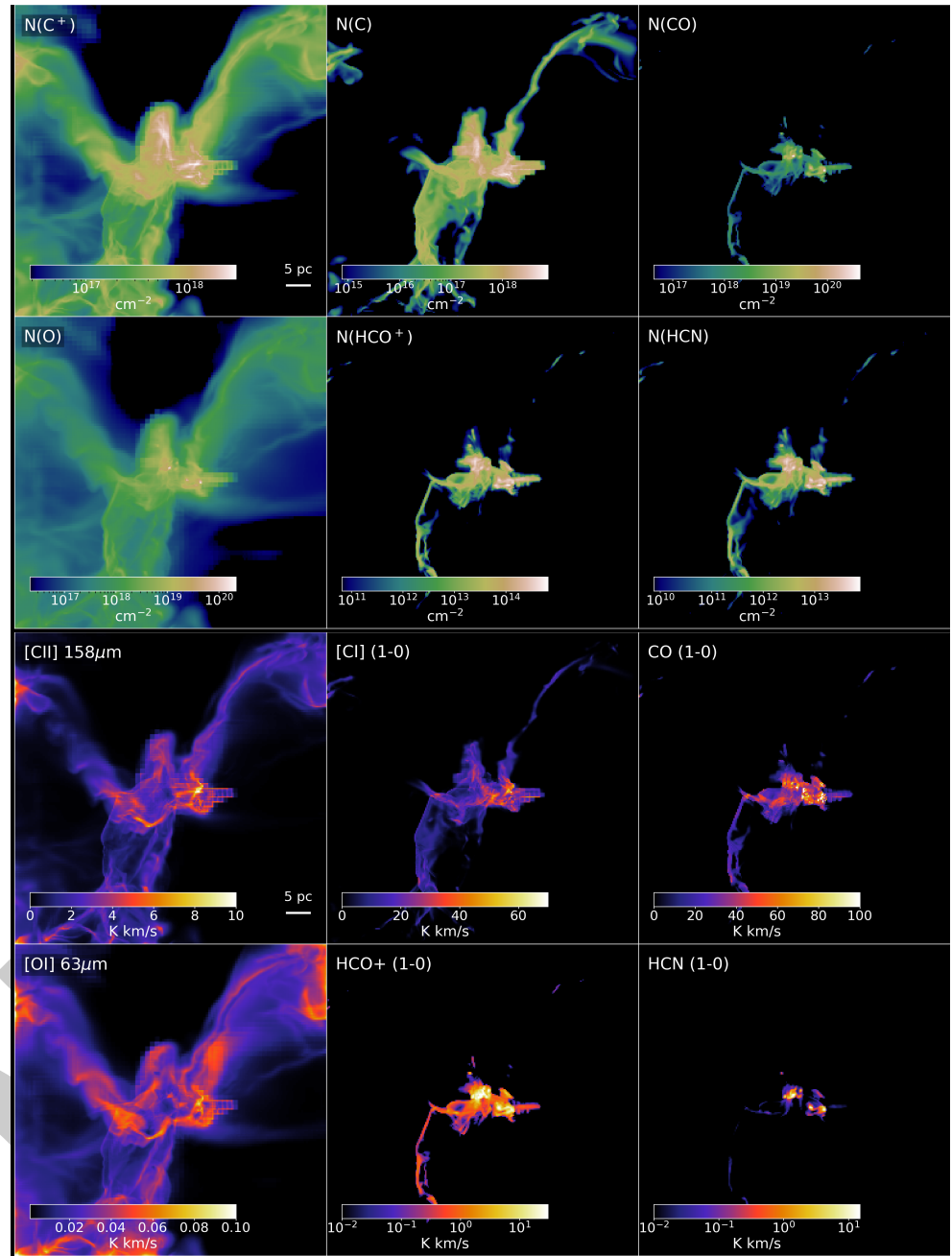


Figure 1: Top half: column densities of C^+ , C and CO (upper row) and O , HCO^+ and HCN (lower row). Bottom half: the corresponding velocity-integrated emission maps, $[CII]$ $158\mu m$, $[CI]$ $(1-0)$ and CO $(1-0)$ (upper row), and $[OI]$ $63\mu m$, HCO^+ $(1-0)$ and HCN $(1-0)$ (lower row). Note that the latter two lines are in log-scale.

125 **Figure 1** shows example outputs of the column densities and velocity integrated emission maps.
 126 The top half shows column densities of C^+ , C and CO (upper row) and O , HCO^+ and HCN
 127 (HCN (lower row). The bottom half shows the corresponding velocity-integrated emission maps, $[CII]$
 128 $158\mu m$, $[CI]$ $(1-0)$ and CO $(1-0)$ (upper row), and $[OI]$ $63\mu m$, HCO^+ $(1-0)$ and HCN $(1-0)$
 129 (HCN (lower row). Note that the colour bars of the latter two velocity integrated emission maps are
 130 in log-scale.

AI usage disclosure

No generative AI tools were used in the development of this software, the writing of this manuscript, or the preparation of supporting materials.

Acknowledgements

This work is supported by the Leading Innovation and Entrepreneurship Team of Zhejiang Province of China (Grant No. 2023R01008). BALG is supported by the German Research Foundation (DFG) in the form of an Emmy Noether Research Group - DFG project #542802847 (GA 3170/3-1).

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